

Fidelity Assessment of Large Language Models across Cross-domain Benchmarks

Seunguk Yu, Hayeong Ryu, Seunghoo Lee and Youngbin Kim

Chung-Ang University, Seoul, Republic of Korea
seungukyu@gmail.com, {bluebarry37, hahaha1321, ybkim85}@cau.ac.kr

Abstract. As LLMs continue to evolve, their ability across domains has become a critical challenge. In this paper, we present a unified evaluation protocol for assessing domain-specific fidelity, grounded in a definition of **fidelity** hallucinations. We categorize it into two subtypes: **aprioristic** hallucination, which reflects a model’s intrinsic knowledge, and **inferential** hallucination, which arises from the model’s reasoning based on supplementary context. Using a diverse collection of academic corpora, we construct and rigorously validate a QA dataset exploring both types of **fidelity** hallucinations across domains, leveraging several LLMs with over 70 billion parameters. We then conduct our evaluation on smaller, commercially deployed models and uncover performance disparities across domains even within the same model. Through comprehensive quantitative analyses, we pinpoint particular domains where hallucinations tend to be more pronounced, along with the conditions that contribute to their occurrence. Furthermore, we demonstrate that strategic variations in decoding methods can significantly impact **fidelity** hallucinations. These findings highlight the importance of domain-aware evaluation protocols and offer practical guidance for the effective deployment of models in real-world applications.

Keywords: Cross-domain Benchmarks · Fidelity Hallucinations · LLMs.

1 Introduction

The rapid advancement of large language models (LLMs) has led to significant improvements across a wide spectrum of downstream tasks, while also enabling their adaptation to specialized domains [14, 4]. In areas such as law, medicine, and engineering, conventional pre-trained models often struggle to capture specialized knowledge and contextual nuances, revealing constraints in domain-specific adaptation. In this aspect, LLMs trained on massive corpora have shown potential in addressing domain-related tasks, particularly those requiring knowledge-intensive question answering (QA), by leveraging their extensive knowledge representations [3, 8].

In this study, we present a unified evaluation for assessing domain-specific fidelity, referred to as **fidelity** hallucination. To facilitate this, we introduce a cross-domain QA dataset construction that ensures procedural consistency,

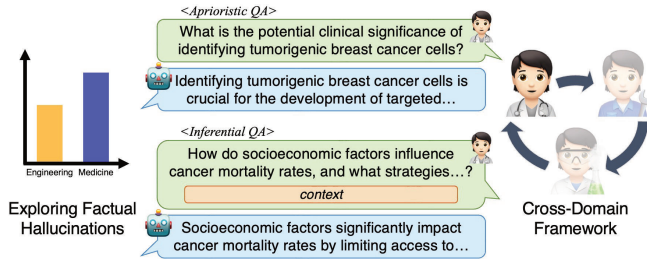


Fig. 1. We evaluated the models with a QA dataset constructed across multiple domains, targeting two distinct subtypes: **aprioristic**, assessing intrinsic knowledge, and **inferential**, evaluating context-based reasoning. The model is required to answer solely from the provided input, without access to the source article.

while capturing the diversity inherent to each *field*¹. This approach is also readily extensible to papers in other domains. We curate research articles by selecting the most highly cited papers to reflect the central themes and dominant research directions. We categorize each QA pair as either **aprioristic** or **inferential** type, based on the nature of knowledge required to generate the answer. The cross-domain evaluation framework with our hallucination types is illustrated in Figure 1. This distinction offers a framework for quantitatively evaluating LLMs’ domain-specific knowledge and their reasoning capabilities.

We generated and verified QA pairs using three LLMs, exceeding 70 billion parameters. These pairs were validated through human annotation, confirming that each of the QA pairs adhered to the given instructions and accurately captured the structural and rhetorical characteristics of the source papers. By applying a standardized data generation and validation protocol across multiple *fields*, we curated a high-fidelity QA dataset reflecting diverse domain knowledge. Leveraging this dataset, we conducted a comprehensive evaluation of the cross-domain generalization capabilities of smaller-scale language models with 1–14 billion parameters, which are more commonly deployed in both academic research and real-world applications.

Our evaluation revealed that the performance when utilizing both internal knowledge and external context varies significantly across *fields*. We also observed that some smaller models demonstrated more stable cross-domain performance, highlighting that larger parameter counts do not necessarily guarantee superior outcomes in domain-specific tasks. These observations emphasize the importance of cross-domain benchmarking when selecting models for specialized applications, where computational efficiency is a key consideration. We also analyzed how different models respond to variations in sampling strategies. Although the dataset and evaluation were conducted on a limited set of *fields* due to space constraints, we plan to release our dataset, validation protocols, and evaluation framework to enable broader adoption.

¹ The distinction between *domain* and *field* in our study is provided in the § 4.1.

2 Related Work

2.1 Domain Shifts in QA Benchmarks

Early benchmarks such as SQuAD and CommonsenseQA focused on extracting explicit information from Wikipedia or passage-based documents [24, 20]. These datasets are often limited by short contextual spans and a restricted diversity of question types. Subsequent efforts have addressed these points by involving more diverse reasoning forms. CoLoTa is designed to assess a model’s generalization on low-frequency commonsense concepts [25]. The task has become increasingly sophisticated by evaluating multi-sentence inference [15], assessing causality in a hierarchical manner [6], and requiring the integration of information distributed across multiple documents [27], thereby moving beyond simple answer forms toward complex reasoning and information synthesis.

With the emergence of LLMs, domain-specific QA has been a key challenge for practical applications [2]. It requires an understanding of domain-specific terminologies and the interpretation of context unique to each domain. While domain-specific benchmarks such as BioASQ, which was introduced for the medical domain [18], and PeerQA covered multiple scientific fields [3], these efforts often focused on narrow subdomains rather than broader categories. More recently, benchmarks such as ExpertQA, SciDQA, and Loong have attempted to expand coverage beyond single domains and incorporate elements such as rationale generation [16, 23]. However, they rely on relatively simple question formats, typically single-sentence queries. We emphasize the need for more structured and contextually rich question formats that can be applied across diverse domains.

2.2 Revisiting Hallucination in Domain Contexts

Hallucination refers to model-generated content that is factually incorrect or unsupported by the source evidence [26]. Recent studies have moved beyond treating hallucination as a surface-level failure, instead proposing new frameworks that reflect the structural aspects of outputs. For instance, in retrieval-augmented generation-based QA tasks, models ignore available supporting evidence when generating answers [21], while in machine translation, recurring but unnecessary patterns have been redefined as structurally motivated hallucinations rather than simple stylistic errors [10].

Hallucination has also been defined within specific domain contexts. In the biomedical domain, long-form responses have been decomposed into atomic information units to identify hallucinations at a fine-grained level [22]. In the legal domain, hallucination patterns have been investigated, such as fictitious case citations that appear authoritative but lack legal grounding [7]. These domain-specific approaches highlight that hallucination cannot be universally defined by a single criterion; instead, its nature often depends on the contextual nuances and inference requirements of the target domain. In this perspective, we propose a framework for evaluating cross-domain QA by categorizing **fidelity** hallucinations into two subtypes: knowledge and reasoning deficiency, enabling a more granular assessment of a model’s domain-specific performance.

3 Preliminaries

3.1 Aprioristic and Inferential Hallucinations

First, **aprioristic** hallucination refers to cases where the absence of inherent knowledge encoded within the model. This type is evaluated by withholding the source text of a question (e.g., a scientific article) and assessing the model’s ability to independently infer domain knowledge based solely on its pre-trained parameters through zero-shot question answering [13].

Inferential hallucination arises from the model’s insufficient reasoning ability from the context. Similarly, this type does not use the source text, and it goes beyond simple question-only settings by examining whether the model can infer answers through reasoning based on the given context. We refer to benchmarks incorporating supplementary passages for reasoning assessment in LLMs [12].

3.2 Goal of Defining Fidelity Hallucinations

An ideal language model is expected to demonstrate extensive knowledge and robust reasoning capabilities. However, in practice, models often exhibit such behaviors: some possess rich intrinsic domain knowledge but show limitations in reasoning with additional context, and vice versa. While no absolute criterion exists to definitively determine a model’s underperformance in a specific domain, our objective is to conduct a comparative analysis across multiple models to identify variations in their capabilities across domains².

4 Cross-domain Dataset Construction

4.1 Research Articles Collection

We utilized OpenAlex³ to access a comprehensive research classification scheme and metadata for each publication. This taxonomy consists of *fields*, with all publications being classified into four broad *domains*. To attain the intended level of granularity, we categorized all publications at the *field*-level. Based on this, we selected (1) *Biochemistry, Genetics & Molecular Biology (Biochemistry)*, (2) *Engineering*, (3) *Medicine*, and (4) *Social Sciences (Social)* as representative *fields* in our dataset construction. We collected approximately 80 papers for each *field*, and converted PDF into markdown format using LlamaParser⁴⁵.

² We evaluated *predicted texts* from the commercial models against *gold texts* from the proposed cross-domain dataset, which was carefully constructed using three models exceeding 70 billion parameters, as detailed later.

³ <https://openalex.org/domains/>

⁴ <https://docs.llamaindex.ai/>

⁵ Our proposed cross-domain dataset construction is designed to be easily extensible to other *fields*.

4.2 Cross-domain QA Generation

We selected three LLMs that demonstrate strong generalization across diverse domains to construct a high-quality QA dataset applicable across research *fields*. Specifically, we employed Llama-3.3-70B-Instruct, Qwen2.5-72B-Instruct, and DeepSeek-R1-Distill-LLaMA-70B for QA generation⁶. These models were chosen based on their large parameter scales, advanced instruction-following abilities, and strong empirical performance across standardized benchmarks [1].

For **aprioristic** QA, the models generated Q_{ap} and A_{ap} that capture the key concepts and contributions of the source article. The questions were instructed to avoid direct reference to specific content (e.g., particular sections). Additionally, they were required to generate rationale R_{ap} to justify why the generated question would be important within the corresponding *field* [17], ensuring that the QA pairs are grounded in domain relevance. Each model generated five pairs for each input, and the QA pairs by the n -th model are as follows:

$$\left\{ (Q, R, A)_{\text{ap}}^{(n,m)} \right\}_{m=1}^5 = \text{LLM}^{(n)}(\text{paper}, \text{prompt}_{\text{ap}}), \quad (1)$$

For **inferential** QA, we intended a more complex reasoning process by incorporating contextual information C . The models were instructed to formulate questions that require additional contexts (e.g., multiple sections) such that the corresponding answers can be inferred through joint consideration of the question and context. We also instructed to self-evaluate candidate questions and generate a rationale R_{in} that supports their choice [17]. Each model generated three pairs for each input, and the QA pairs by the n -th model are as follows:

$$\left\{ (Q, C, R, A)_{\text{in}}^{(n,m)} \right\}_{m=1}^3 = \text{LLM}^{(n)}(\text{paper}, \text{prompt}_{\text{in}}), \quad (2)$$

4.3 Cross-domain QA Verification

To validate whether generated QA pairs could be justified from the perspectives of other models, we employed natural language inference (NLI), a reasoning task that determines whether a given *premise* logically entails a *hypothesis* [5]. We generated multiple QA pairs per paper to prevent the questions from being biased toward specific content. We assumed that if a QA pair generated by a model could also be justified by rationales by other models, then the <entailment> score between them would be high. We treated the QA pair generated by a specific model as a *premise* and each rationale generated by different models as a *hypothesis* to compute the probability of the <entailment> label.

For the m' -th QA pair generated by the n' -th model, we computed the <entailment> label probability prob_{ap} between this pair and each rationale generated by all other models (excluding the n' -th model). The *inter-score* for the

⁶ We conducted human validation to assess the instruction-faithfulness of the generated QA pairs, and observed high consistency from the human raters.

aprioristic QA is then defined as the average of these probabilities, as follows:

$$\text{prob}_{\text{ap}}^{(n',m'),(n,m)} = \text{NLI} \left((Q, A)_{\text{ap}}^{(n',m')}, R_{\text{ap}}^{(n,m)} \right), \quad (3)$$

$$\text{inter-score}_{\text{ap}}^{(n',m')} = \text{avg} \left(\text{prob}_{\text{ap}}^{(n',m'),(n,m)} \right), \quad (4)$$

where $n, n' \in [1, 3]$, $m, m' \in [1, 5]$, $n' \neq n$

When applied to **inferential** QA, the context C associated with each pair is additionally concatenated with the *premise*. Accordingly, from the initial set of 15 **aprioristic** and 9 **inferential** QA pairs per paper, we selected the top 3 and 2 pairs respectively with the highest *inter-score* for each type, resulting in a final dataset of 1,600 QA instances. To mitigate potential biases that may arise from relying on a single model, even when utilizing large-scale models with over 70 billion parameters, we employed each model to generate multiple QA pairs and validated them using other models in a cross-verification setting. This approach integrates diverse generative behaviors at the dataset level, improving cross-model generalization in domain-specific contexts.

5 Cross-domain Evaluations

5.1 Experimental Settings

To ensure coverage across diverse architectures, we selected the recently released models for evaluation, including the Llama-3.2-Instruct, Qwen-3, and DeepSeek-R1-Distill series⁷. We conducted multiple runs for each model under three temperature values: 0, 0.3, and 0.6, enabling a systematic analysis of the consistency and variability of model responses.

5.2 Fidelity Metric

We regarded the answers from the dataset as *gold text* G , and responses generated by smaller models as *predicted text* P . We assessed two aspects: first, whether the textual forms of P and G are similar in terms of lexical similarity, for word usage and overall expressions; second, whether the semantic similarity conveyed by both texts is consistent. The former was measured using BLEU, whereas the latter was evaluated based on semantic similarity by BERTScore, with FacebookAI/roberta-large as the underlying model⁸. We integrated these metrics into a single F1 score form to provide a unified assessment as follows:

$$\text{score}_{\text{ap,in}} = \frac{2 * \text{BLEU}(G, P) * \text{BertScore}(G, P)}{\text{BLEU}(G, P) + \text{BertScore}(G, P)}. \quad (5)$$

⁷ For models such as DeepSeek and Qwen that utilize a <think> token for reasoning, final answers were derived through a post-processing stage.

⁸ We use 1-gram BLEU to capture lexical tendencies rather than more exact matches such as 4-gram, as models weren't trained to reproduce identical outputs. We also utilize BERTScore to evaluate semantic similarity to complement this aspect.

Table 1. We report the average score over multiple runs with varying decoding temperatures. The values in parentheses indicate the differences observed when the question type was shifted from **aprioristic** to **inferential**. The **best-performing** and **second-best** models in each *field* are highlighted for comparison.

<i>Fields</i>	<i>Biochemistry</i>			<i>Engineering</i>		
	score _{ap}	score _{in}	Avg	score _{ap}	score _{in}	Avg
Fidelity Metrics						
Llama-3.2-1B-Instruct	25.51	39.26 (+13.75)	32.38	27.98	40.01 (+12.03)	33.99
Llama-3.2-3B-Instruct	28.77	41.01 (+12.24)	34.89	30.63	43.50 (+12.86)	37.06
Qwen3-1.7B	27.40	42.80 (+15.39)	35.10	28.80	41.47 (+12.66)	35.13
Qwen3-4B	28.85	40.13 (+11.28)	34.49	28.66	39.69 (+11.03)	34.18
Qwen3-8B	29.24	39.79 (+10.55)	34.52	29.83	39.12 (+ 9.29)	34.48
Qwen3-14B	30.56	39.87 (+ 9.31)	35.22	31.12	39.56 (+ 8.44)	35.34
DeepSeek-R1-Distill-Llama-8B	26.48	44.09 (+17.61)	35.28	25.70	43.20 (+17.50)	34.45
DeepSeek-R1-Distill-Qwen-1.5B	21.91	39.61 (+17.70)	30.76	24.70	38.48 (+13.78)	31.59
DeepSeek-R1-Distill-Qwen-7B	24.97	41.05 (+16.07)	33.01	24.69	40.09 (+15.39)	32.39
DeepSeek-R1-Distill-Qwen-14B	29.76	45.01 (+15.25)	37.39	29.85	45.89 (+16.03)	37.87
Gemma-3-1b-it	26.94	38.32 (+11.38)	32.63	27.41	38.41 (+10.99)	32.91
Gemma-3-4b-it	25.03	33.94 (+ 8.91)	29.48	25.88	34.71 (+ 8.83)	30.29
Gemma-3-12b-it	28.01	37.18 (+ 9.17)	32.60	27.71	37.32 (+ 9.60)	32.52

<i>Fields</i>	<i>Medicine</i>			<i>Social</i>		
	score _{ap}	score _{in}	Avg	score _{ap}	score _{in}	Avg
Fidelity Metrics						
Llama-3.2-1B-Instruct	27.42	38.45 (+11.02)	32.93	26.93	37.83 (+10.90)	32.38
Llama-3.2-3B-Instruct	31.12	41.77 (+10.65)	36.44	29.99	40.57 (+10.57)	35.28
Qwen3-1.7B	32.64	43.58 (+10.94)	38.11	31.44	39.08 (+ 7.63)	35.26
Qwen3-4B	30.12	39.76 (+ 9.64)	34.94	28.74	36.35 (+ 7.61)	32.55
Qwen3-8B	31.11	39.31 (+ 8.20)	35.21	28.83	35.62 (+ 6.78)	32.23
Qwen3-14B	31.48	40.02 (+ 8.53)	35.75	29.76	36.90 (+ 7.13)	33.33
DeepSeek-R1-Distill-Llama-8B	27.39	44.49 (+17.09)	35.94	25.74	40.65 (+14.91)	33.19
DeepSeek-R1-Distill-Qwen-1.5B	20.48	37.08 (+16.59)	28.78	21.16	37.50 (+16.34)	29.33
DeepSeek-R1-Distill-Qwen-7B	24.14	41.20 (+17.05)	32.67	24.64	39.11 (+14.46)	31.87
DeepSeek-R1-Distill-Qwen-14B	29.91	45.99 (+16.08)	37.95	28.97	42.82 (+13.85)	35.89
Gemma-3-1b-it	28.23	37.97 (+ 9.74)	33.10	26.77	35.59 (+ 8.81)	31.18
Gemma-3-4b-it	27.01	35.56 (+ 8.54)	31.29	25.50	31.93 (+ 6.42)	28.72
Gemma-3-12b-it	29.25	38.26 (+ 8.31)	34.11	27.63	34.78 (+ 7.15)	31.20

5.3 Experimental Results

Main Results We present the results of **fidelity** hallucination evaluations for the selected models in Table 1. Although prior research has established a correlation between model size and task performance [11], we demonstrate that in domain-specific scenarios, careful selection of model type and parameter size can achieve competitive performance with significantly reduced computational overhead. We reveal that DeepSeek-R1-Distill-Qwen-14B consistently ranked among the top two performers across the four *fields*. Among other models with strong average performance, the leading candidates varied depending on the *fields*: DeepSeek-R1-Distill-Llama-8B achieved the highest performance in *Biochemistry*, Llama-3.2-3B-Instruct outperformed others in *Engineering* and *Social*, and Qwen3-1.7B demonstrated superior results in *Medicine*.

Analysis in Fidelity Hallucination Types The **inferential** QA consistently yields superior results compared to the **aprioristic** QA, as shown in Table 1. These findings suggest that models leveraged contextual information to enhance inference in domain texts. We found that the DeepSeek-R1-Distill series

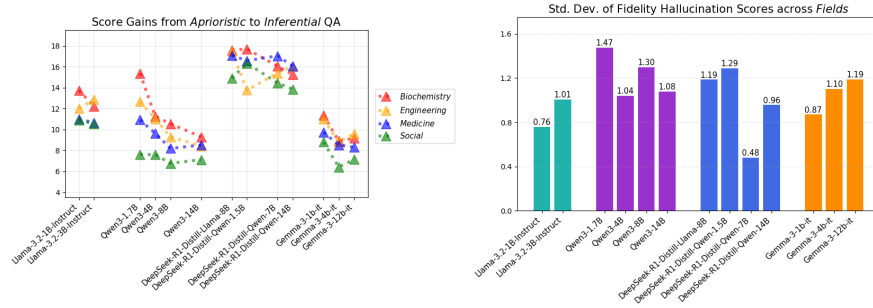


Fig. 2. (Left) Score gain comparisons from **aprioristic** to **inferential**. We use the average score over multiple runs with varying decoding parameters. (Right) Standard deviation comparisons of average **fidelity** hallucination scores. For each model, we report the average score over multiple runs with varying decoding temperatures.

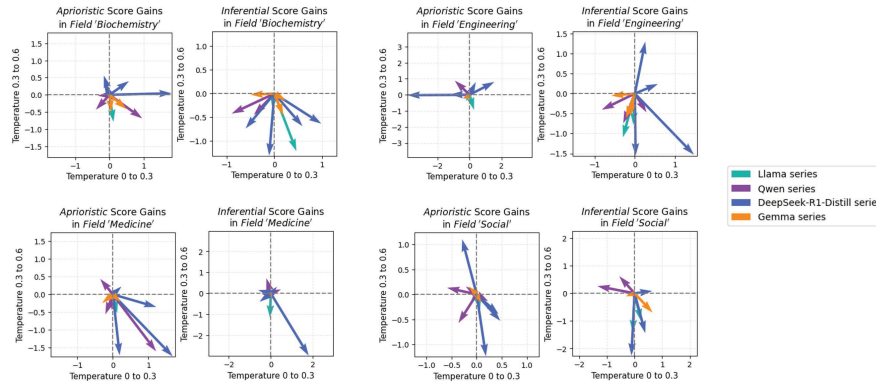


Fig. 3. Score gain variations as a function of temperature are reported, stratified by both *field* and type of **fidelity** hallucinations. For each *field*, the left/right panel depicts performance changes under the **aprioristic**/**inferential** QA.

exhibited substantial score improvements across models. This trend stems from distillation from teacher models with over 600B parameters, which may confer a stronger ability to perform context-driven inference. We provide the score gain details for each *field* in Figure 2 (Left). Notably, in *Biochemistry*, all models exhibited markedly greater score gains than those in other *fields*. Moreover, smaller-scale models within each series often achieved larger improvements than their larger models, suggesting that models with limited capacity can sometimes outperform larger ones when supported by external context.

Analysis in Cross-domain Consistency To quantify the variability of model performance, we computed the standard deviation of *field*-specific average scores for each model, as presented in Figure 2 (Right). The DeepSeek-R1-Distill-Qwen-7B and LLaMA-3.2-1B-Instruct exhibited the lowest standard deviations

Table 2. We report the average (standard deviation) of scores according to top-p and top-k sampling, while fixing the temperature at 0.6. The **best-performing** and **second-best** models in each setting are highlighted for comparison.

Fidelity Hallucinations Hyperparams	<i>Field</i>		<i>Top-p Sampling</i>				<i>Top-k Sampling</i>			
			Aprioristic		Inferential		Aprioristic		Inferential	
			p=0.7	p=0.8	p=0.7	p=0.8	k=15	k=25	k=15	k=25
Llama-3.2-1B-Instruct			0.46 (0.17)	0.45 (0.39)	0.90 (0.30)	0.67 (0.41)	-0.10 (0.26)	-0.10 (0.31)	-0.08 (0.23)	0.15 (0.38)
Llama-3.2-3B-Instruct			0.33 (0.43)	0.26 (0.53)	0.90 (0.60)	0.61 (0.42)	0.15 (0.37)	0.03 (0.32)	0.04 (0.33)	-0.28 (0.35)
Qwen3-1.7B			2.55 (1.35)	2.21 (1.46)	0.03 (0.30)	0.05 (0.44)	2.32 (1.53)	2.10 (1.35)	-0.05 (0.48)	-0.01 (0.51)
Qwen3-4B			1.74 (0.43)	1.65 (0.45)	0.25 (0.37)	0.10 (0.42)	1.88 (0.55)	1.86 (0.26)	-0.01 (0.72)	-0.09 (0.54)
Qwen3-8B			1.58 (0.27)	1.45 (0.32)	0.11 (0.30)	-0.34 (0.48)	1.41 (0.37)	1.35 (0.58)	0.14 (0.17)	0.08 (0.27)
Qwen3-14B			0.46 (0.32)	0.39 (0.45)	0.01 (0.47)	-0.11 (0.25)	0.40 (0.41)	0.39 (0.43)	0.23 (0.44)	0.15 (0.34)
DeepSeek-R1-Distill-Llama-8B			5.57 (0.30)	5.15 (0.62)	0.90 (0.35)	0.29 (0.35)	4.73 (0.42)	5.58 (0.93)	-0.14 (0.44)	0.38 (0.53)
DeepSeek-R1-Distill-Qwen-1.5B			5.55 (0.28)	4.56 (0.84)	0.96 (0.99)	-1.04 (0.95)	4.99 (0.80)	4.81 (0.79)	0.26 (1.20)	0.60 (0.78)
DeepSeek-R1-Distill-Qwen-7B			4.95 (0.92)	4.58 (0.78)	-0.67 (0.22)	-1.17 (0.58)	5.10 (0.62)	5.30 (0.56)	0.27 (0.41)	-0.08 (0.31)
DeepSeek-R1-Distill-Qwen-14B			3.86 (1.54)	3.55 (1.51)	0.22 (0.56)	-0.32 (0.67)	3.65 (1.78)	3.84 (1.62)	-0.23 (0.20)	-1.12 (0.81)
Gemma-3-1b-it			0.29 (0.20)	0.15 (0.10)	0.35 (0.30)	0.05 (0.15)	0.13 (0.18)	0.15 (0.15)	0.16 (0.25)	0.05 (0.32)
Gemma-3-4b-it			0.25 (0.21)	0.03 (0.24)	0.11 (0.32)	0.11 (0.17)	-0.06 (0.19)	0.001 (0.11)	0.03 (0.18)	0.12 (0.37)
Gemma-3-12b-it			0.13 (0.28)	0.11 (0.19)	0.04 (0.34)	0.14 (0.18)	-0.15 (0.07)	-0.12 (0.26)	0.09 (0.30)	0.002 (0.42)

0.48 and 0.76, suggesting relatively high consistency. However, their average fidelity hallucination scores remained lower than those of other models. This highlights the inherent difficulty in achieving both cross-domain consistency and high domain-specific performance within a single model. Real-world domain adaptation is costly in terms of data curation and training stability. Therefore, cross-domain evaluation is crucial for maximizing the utility of pre-trained models without domain-specific fine-tuning.

5.4 Temperature-controlled Decoding Setup

We present Figure 3 to illustrate the sensitivity of model scores to temperature variation in greater detail: the x and y axes denote the change from temperature 0.0 to 0.3 and 0.3 to 0.6⁹. We found that model score did not exhibit a uniform trend in this regard. Although the direction of performance shifts varied across *fields*, DeepSeek-R1-Distill series showed that the magnitude of fluctuation was consistently larger than other models. Such behavior raises concerns regarding the cross-domain robustness of models that are overly sensitive to sampling temperature [9]. Conversely, the Gemma series demonstrated minimal performance variation across temperature settings, indicating a more stable encoding of domain-generalizable knowledge regardless of decoding stochasticity.

5.5 Sampling Strategies on Cross-domain Analysis

We extend our analysis beyond sampling-based decoding to incorporate top-p and top-k sampling strategies with the considered hyperparameters from prior research [19]. We quantified score changes in each strategy as in Table 2, computing the mean and standard deviation of the results. The DeepSeek-R1-Distill

⁹ Models appearing in the first quadrant exhibit benefits from stochastic decoding. In contrast, models in the third quadrant perform best under greedy decoding, indicating degradation with higher sampling entropy.

series exhibited particularly pronounced fluctuations in **aprioristic** QA. This suggests that strategic token selection during decoding can effectively leverage distilled knowledge to produce domain-specific information. Notably, these score gains did not correlate with model size as the smallest model (Qwen3-1.7B) showed the most significant improvement. However, for **inferential** QA, sampling strategies yielded limited improvements. This suggests that the token-level sampling variability during decoding may interfere with the model’s ability to maintain coherent contextual inferences.

6 Conclusion

We propose a unified evaluation framework for domain-specific texts, named **fidelity** hallucinations, including **aprioristic** and **inferential** QA. We constructed a cross-domain dataset by leveraging large-scale models. Our evaluation revealed that cross-domain performance does not strictly correlate with model size. Notably, smaller models sometimes outperform larger ones when leveraging external context, as in **inferential** QA. We also analyzed cross-domain consistency and sampling strategies, which highlighted the necessity of domain-aware evaluation frameworks. These results underscore the need for more strategic cross-domain evaluation beyond model size for effective LLM deployment.

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